Technical evidence ledger

Metrics, assumptions, limitations and traceability

This appendix contains the technical detail intentionally removed from the public competition report.

CUSTOM HOLOOCEAN	**vehicle + native sonar + safety**	not the reconstruction headline
FLAGSHIP 2-D	**demo-optimised analytic surrogate**	not field or CFD truth
EQUAL-EVIDENCE TEST	**relative benchmark under fixed budget**	not universal superiority
VOLUMETRIC MISSION	**one anisotropic 3-D GP**	not measured environmental volume

STATUS: INITIAL PROTOTYPE OR MODEL | EVIDENCE FREEZE: 22 JULY 2026

01 / IN-ENGINE EVIDENCE

Isolation, mission and localisation evidence

ISOLATION

- Mission instance bwp26-fa3; cinematic instance bwp26-cin1.
- Private shared memory, semaphores, instance IDs, octrees, caches, temp, logs and outputs.
- Both manifests record launcher-owned engine PIDs and zero external process termination.

CUSTOM-HOLOOCEAN MISSION

- 564 simulated CT readings; 394.85 m travelled.
- 0 collisions; 4 detours; 1.85 m minimum measured clearance vs 2.0 m target.
- REVIEW against non-compliant truth; RMSE 2.7203 PSU; F1 and IoU 0.000.
- Use as integration and safety evidence, not reconstruction evidence.

Forward-looking imaging-sonar localisation

2.353 m
centre error

1.669 m
posterior radius

4.771 m
confirmation spread

5 / 5
valid non-fallback prior trials

Two independent search radii; inverse-uncertainty consensus; 5.0 m spread limit.

No oracle input and no silent fallback. Truth entered only after estimation for scoring.

02 / FLAGSHIP + SCREENING

Flagship performance and all three screen states

0.3415 PSU
plume RMSE

0.947
boundary F1

0.900
boundary IoU

POSSIBLE EXCEEDANCE
correct + conclusive

Flagship: 591 samples; 516.55 / 520 m; coverage 0.770; P(exceed) 1.000; +1.234 PSU maximum reconstructed exceedance.

Case	Truth	Output	Samples	P(exceed)	RMSE	F1
Compliant reference	PASS	CLEAR	1,049	0.036	0.0487	1.000
Borderline	PASS	REVIEW	290	0.363	0.5749	0.783
Flagship	FAIL	POSSIBLE EXCEEDANCE	591	1.000	0.3415	0.947

SCENARIO DISCLOSURE

The flagship intentionally uses a high-contrast analytic plume surrogate so the workflow is understandable. It preserves the accepted outfall geometry and is not field or CFD truth.

03 / EQUAL-EVIDENCE BENCHMARK

Complete metric and screening table

Eight seeds per method; 48 readings; 300 m cap; same area, analytic plume and 0.03 PSU sensor noise.

Metric	Sparse fixed	Regular	Adaptive
Plume RMSE (PSU)	1.151	1.661	0.758
Boundary F1	0.000	0.287	0.789
Boundary IoU	0.000	0.177	0.656
Missed-plume fraction	1.000	0.817	0.284
Useful-sample fraction	0.167	0.229	0.701
Mean posterior std (PSU)	2.169	2.359	2.292
Max outside std (PSU)	2.877	4.500	4.496
Travel distance (m)	297.8	300.0	297.5
Indicative time (min)	44.27	9.93	9.86

Screening counts

Method	CLEAR	POSSIBLE	REVIEW	Conclusive	Accuracy	False C / E
Sparse fixed	0/8	0/8	8/8	0%	n/a	0 / 0
Regular	0/8	4/8	4/8	50%	100%	0 / 0
Adaptive	0/8	8/8	0/8	100%	100%	0 / 0
All	0/24	12/24	12/24	50%	100%	0 / 0

Sparse design: 24 fixed stations plus one replicate each. REVIEW is not a miss, but it is not a conclusive detection; both rates remain visible.

04 / VOLUME + FEASIBILITY

3-D evidence and planning assumptions

VOLUMETRIC MISSION

- Four bands: 0.8, 1.6, 2.8 and 4.5 m; 912 total samples.
- One anisotropic 3-D GP; RMSE 0.477 PSU; MAE 0.238 PSU; volume IoU 0.805.
- 2,051.3 m3 reconstructed vs 2,448.2 m3 surrogate truth: 16.2% underestimation.
- Uncertain volume 24,351.4 m3; mean std inside plume 1.551 PSU.

COST MODEL

- Full build: USD 21.5k-44k.
- Incremental with owned ROV + sonar: USD 9k-22.5k.
- Survey day: shore USD 3.4k-8.9k; small boat USD 4.9k-13.4k.
- Midpoint break-even illustration: 2-7 repeats; not guaranteed.

Assumptions and failure conditions

- Two operators; 2.5-4 h on water; 4-8 h reporting; 12 missions/year; five-year amortisation.
- Public anchors at review: BlueROV2 from USD 4,900; Omniscan 450 FS from USD 2,490. Local CT, integration, vessel, labour and permits require quotes.
- If vessel, permits and accredited sampling remain unchanged, savings may disappear. The value becomes better spatial evidence and repeatable history.

05 / CLAIMS BOUNDARY

What the evidence supports - and what it does not

SUPPORTED

- Simulation-led prototype feasibility.
- Isolated collision-free custom-HoloOcean integration.
- Sonar localisation without oracle input.
- Uncertainty-aware three-state screening.
- Relative adaptive performance in the stated benchmark.
- Coherent 2-D / 3-D surrogate maps and digital-twin concept.

NOT SUPPORTED

- Regulatory certification or autonomous compliance.
- Field accuracy or CFD validation.
- Guaranteed savings or break-even.
- Universal replacement of accredited monitoring.
- Unsupervised field deployment readiness.
- Measured environmental plume volume.

NEXT VALIDATION GATE

Integrate a calibrated conductivity-temperature payload, establish traceable timing and calibration, validate known gradients in controlled water, then run a supervised nearshore pilot with independent reference samples.

Video disclosure: 625 consecutive Full-HD HoloOcean frames; dedicated cinematic path, not a telemetry-synchronised replay; fixed scientific result panels.